

Lecture 9

HHU summer term 2026

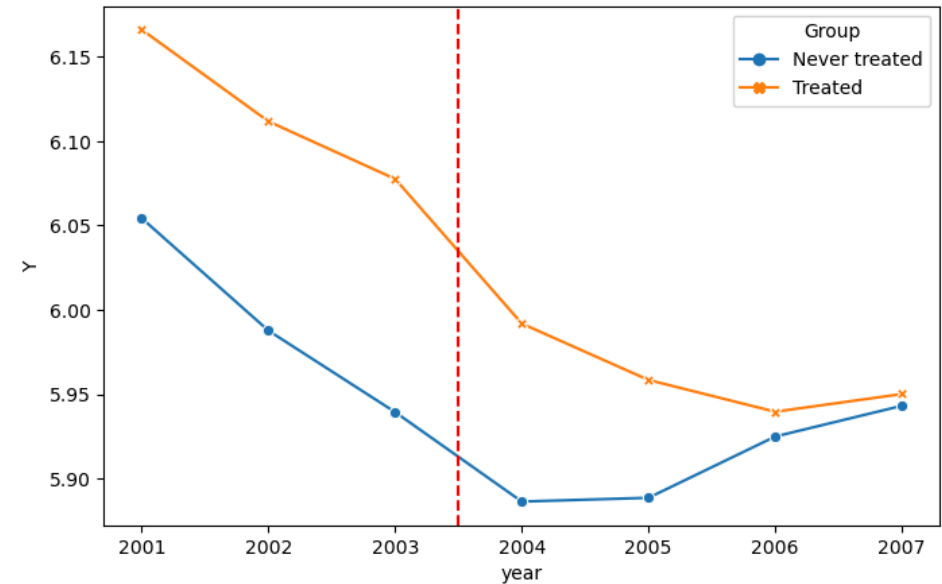
July 22, 2026

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Difference-in-Differences

Motivation

- So far: Cross-sectional data
 - At a given point in time t , we observe n individuals (i.i.d)
- Now: Panel data and repeated cross-sectional data
 - We observe data for individuals at multiple points in time
- How can we estimate a causal effect in such a setting?



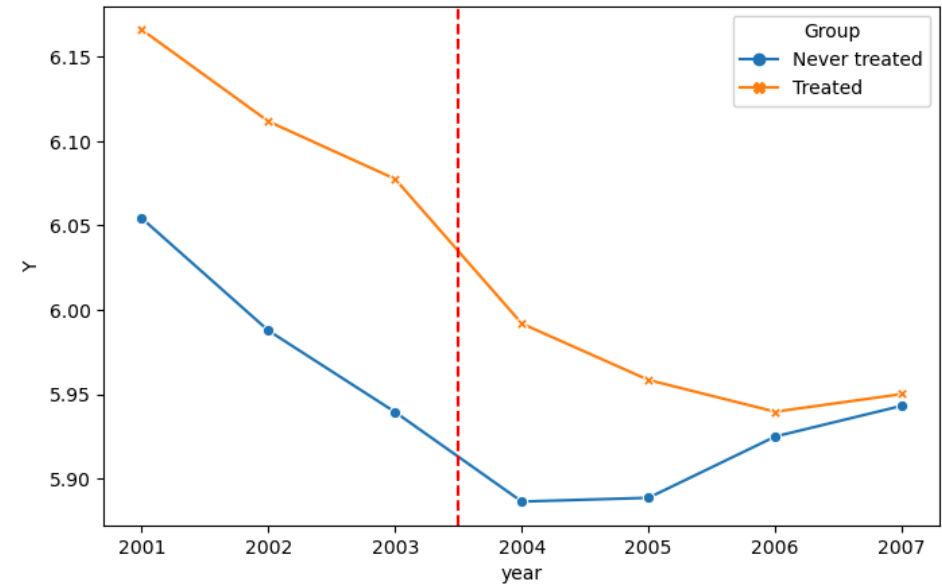
Motivation

- **Difference-in-Differences (DiD)** is one of the most popular identification strategies

Stylized example:

- Consider 2 US states
- One introduced a higher minimum wage in 2003, the other not

*What is the causal effect of a **minimum wage increase** on **youth unemployment**?*



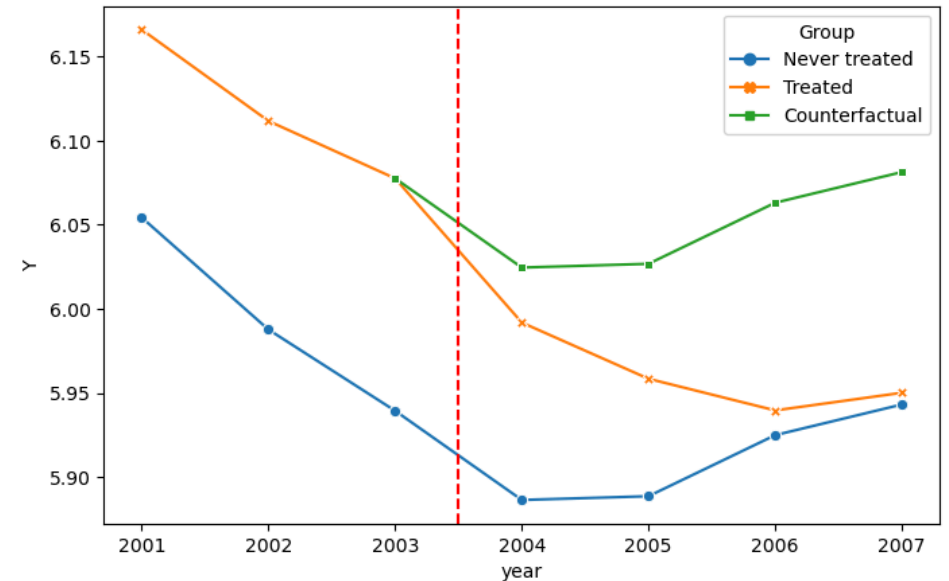
Motivation

- **Difference-in-Differences (DiD)** is one of the most popular identification strategies

Stylized example:

- Consider 2 US states
- One introduced a higher minimum wage in 2003, the other not

*What is the causal effect of a **minimum wage increase** on **youth unemployment**?*



- The underlying assumptions must apply to *what would have happened in the treated state would it not have been treated in 2003*:
Parallel trends

Data Types

Panel Data

Observe a subsample of individuals over a time period

Example:

Business to Business

Repeated Cross Sections

At each time, observe a subsample of individuals

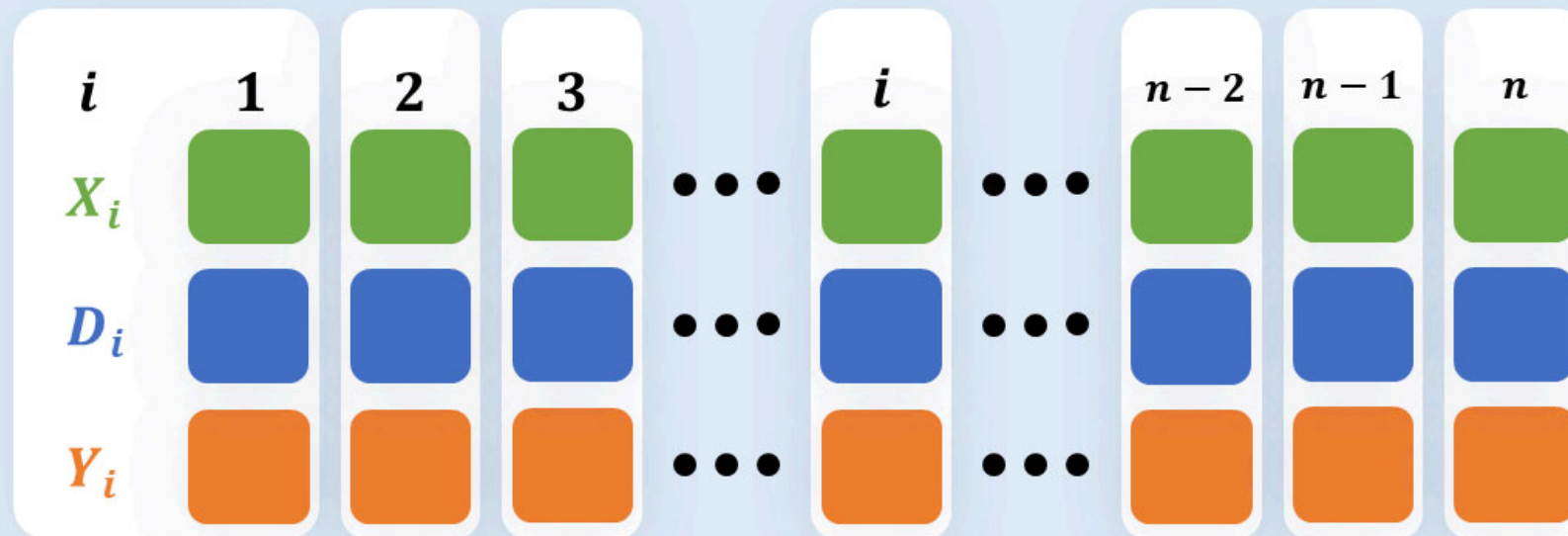
Example:

Business to Consumer

Motivation: Types of Data

Cross-Sectional Data

Example: Observe one random sample of website visitors

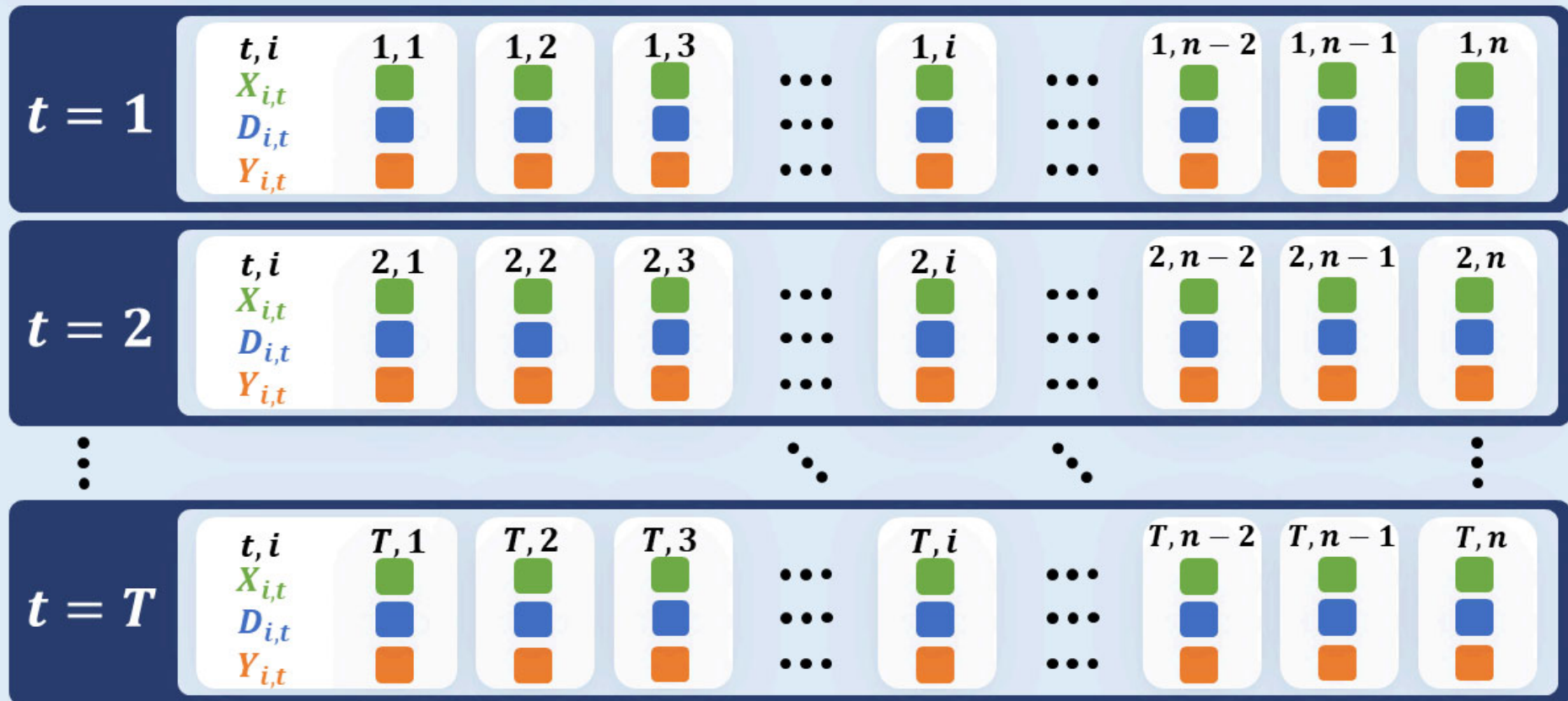


Cross-sectional data

Motivation: Types of Data

Panel Data

Example: Observe (regular) customers at several points in time – B2B

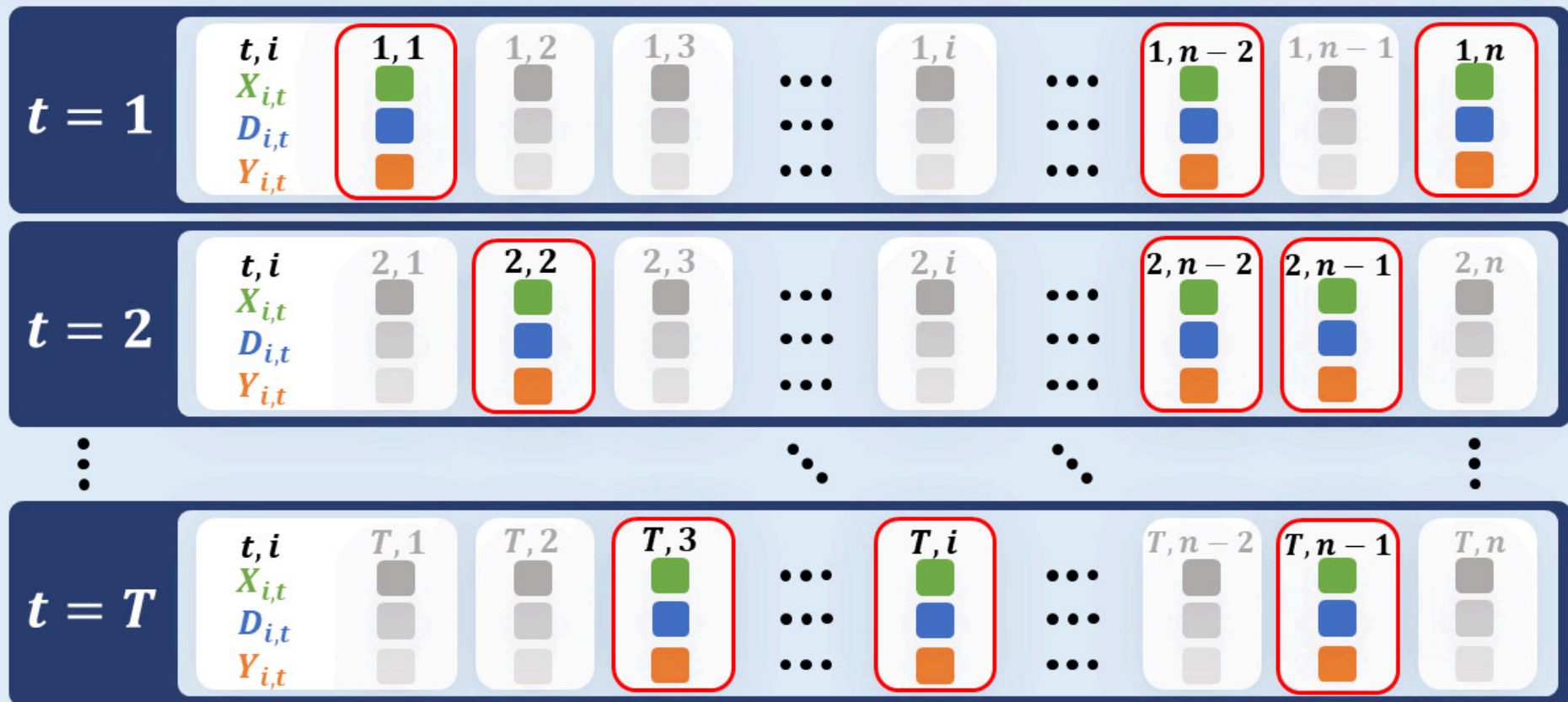


Panel data

Motivation: Types of Data

Repeated Cross-Section

Example: Observe random subset of customers at several points in time (B2C)

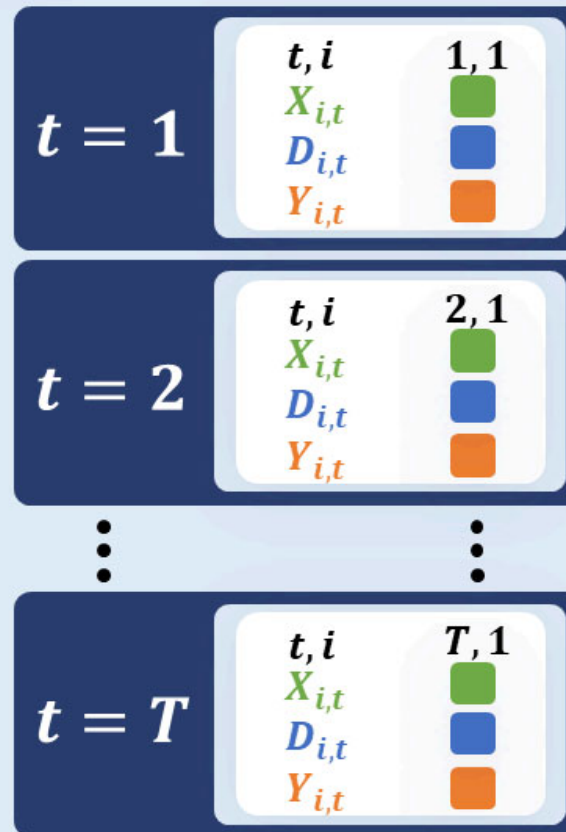


Repeated cross section

Motivation: Types of Data

Time Series

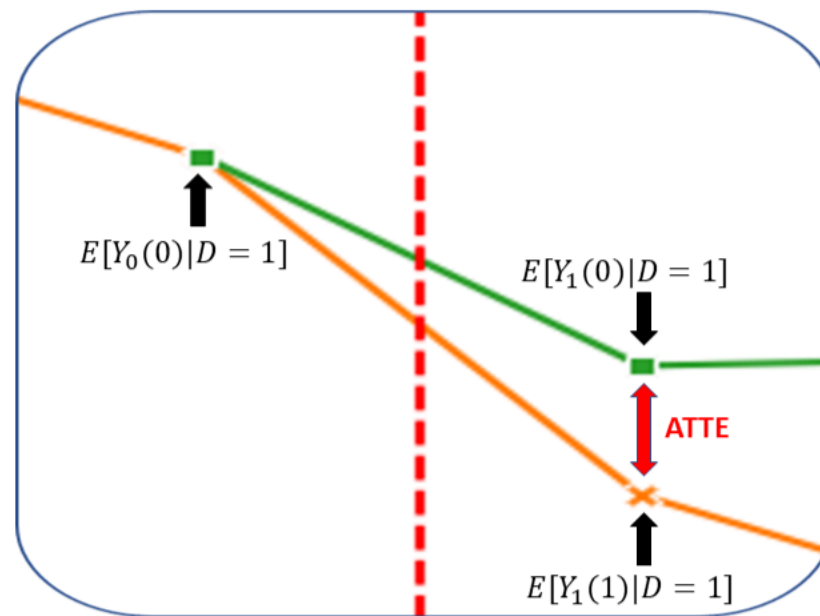
Example: Observe sales in one market over multiple periods



Time series

Setting: 2 Time Periods

- Two time periods: $t = 0$ pre-treatment and $t = 1$ post-treatment
- Y_t : Outcome of interest at time t
- D : Treatment indicator if treated between $t = 0$ and $t = 1$ (binary treatment)
- $Y_t(0)$: Potential outcome of interest at time t if **not** treated up until 1
- $Y_t(1)$: Potential outcome of interest at time t if treated up until 1



Motivation

- The counterfactuals (green) are not observed
- Rely on the group of untreated units (blue) to estimate the counterfactuals

Motivation

Key assumption: Parallel trends

$$\underbrace{\mathbb{E}[Y_1(0)|D=1] - \mathbb{E}[Y_0(0)|D=1]}_{=\Delta_{D=1}(0)} = \underbrace{\mathbb{E}[Y_1(0)|D=0] - \mathbb{E}[Y_0(0)|D=0]}_{=\Delta_{D=0}(0)}$$

Motivation

- The parameter of interest is the **average treatment effect on the treated (ATTE)**:

$$\theta_0 := \mathbb{E}[Y_1(1) - Y_1(0) | D = 1]$$

- If the parallel trends assumptions is satisfied, the **ATTE** can be identified by

$$\theta_0 := \underbrace{\Delta_{D=1}(1)}_{\text{Difference in treated}} - \underbrace{\Delta_{D=0}(0)}_{\text{Difference in untreated}}$$

Extension of the Parallel Trend Assumption

- The parallel trend assumption might be very hard to justify
- Usually there are some previous conditions which influence the trend of the outcome (e.g. economic growth, population, etc.)
- Extend the parallel trend assumption to hold conditional on some covariates X

Conditional parallel trends

$$\mathbb{E}[Y_1(0) - Y_0(0)|X, D = 1] = \mathbb{E}[Y_1(0) - Y_0(0)|X, D = 0]$$

Identifying Assumptions

Conditional parallel trends:

$$\mathbb{E}[Y_1(0) - Y_0(0)|X, D = 1] = \mathbb{E}[Y_1(0) - Y_0(0)|X, D = 0]$$

No anticipation:

$$\mathbb{E}[Y_0(0)|X, D = 1] = \mathbb{E}[Y_0(1)|X, D = 1]$$

Overlap:

$$\exists \epsilon > 0 : P(D = 1|X) > \epsilon \text{ and } P(D = 1|X) \leq 1 - \epsilon$$

Outlook: Extensions

- So far, we considered a setting with 2 periods
- It's possible to extend the idea to settings with multiple periods, see for example Callaway and Sant'Anna (2021)
- Moreover, it is possible to use sensitivity analysis as of Chernozhukov et al. (2022) for Difference-in-Differences
- Overview on Difference-in-Differences models: Chapter 7 of Huber (2023), Chapter 15 in Chernozhukov et al. (forthcoming), Callaway (2022) and Roth et al. (2023) for more recent developments

Panel Data

Panel Data - Observational Setting

- Observations are iid. of the form $W = (Y_0, Y_1, D, X)$
- Score implemented as proposed in Chang (2020), Sant'Anna and Zhao (2020) and Zimmert (2018)

$$\psi(W, \theta, \eta) = -\frac{D}{p}\theta + \frac{D - m(X)}{p(1 - m(X))}(Y_1 - Y_0 - g(0, X))$$

with $\eta = (g, m, p)$.

- Nuisance components

$$\begin{aligned}p_0 &= \mathbb{E}[D] \\m_0(X) &= P(D = 1|X) \\g_0(0, X) &= \mathbb{E}[Y_1 - Y_0|D = 0, X]\end{aligned}$$

Implementation - Data Backend

- Take a look at the data example

► Code

	year	Groups	Y	lpop	lavg_pay	id	D	region_2	region_1
2	2003	Never treated	5.111988	9.790095	10.382265	13001	0	0	1
3	2004	Never treated	5.308268	9.792891	10.424986	13001	0	0	1
9	2003	Never treated	3.401197	8.983314	10.053802	13003	0	0	1

Implementation - Data Backend

- **DoubleMLDIDData** is constructed for a univariate outcome $y \Rightarrow$ Use $\Delta Y = Y_1 - Y_0$
- Choose features as pre-treatment covariates

► Code

	id	lpop	lavg_pay	region_2	region_3	region_4	D	Y
0	13001	9.790095	10.382265	0	1	0	0	0.196280
1	13003	8.983314	10.053802	0	1	0	0	0.154151
2	13005	9.237274	10.059679	0	1	0	0	0.054559
3	13007	8.364042	10.085684	0	1	0	0	-0.262364
4	13009	10.716150	10.168042	0	1	0	0	-0.175585

Implementation - Data Backend

- Construct a basic `DoubleMLDIDData` object

```
1 from doubleml import DoubleMLDIDData
2
3 dml_data = DoubleMLDIDData(df_diff, y_col="Y", d_cols="D",
4                             x_cols=["lpop", "lavg_pay", "region_2", "region_3", "region_4"])
5 print(dml_data)
```

```
===== DoubleMLDIDData Object =====
Time variable: None
Outcome variable: Y
Treatment variable(s): ['D']
Covariates: ['lpop', 'lavg_pay', 'region_2', 'region_3', 'region_4']
Instrument variable(s): None
No. Observations: 1519
```

Implementation - Model Class

- Implemented via the DoubleMLDID class

```
1 from doubleml import DoubleMLDID
2 from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
3 from sklearn.base import clone
4
5 ml_g = RandomForestRegressor()
6 ml_m = RandomForestClassifier()
7
8 dml_did = DoubleMLDID(dml_data,
9                       ml_g=clone(ml_g),
10                      ml_m=clone(ml_m))
```

Implementation - Estimation

- Estimation via the `fit()` method

```
1 from doubleml import DoubleMLDID
2 from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
3 from sklearn.base import clone
4
5 ml_g = RandomForestRegressor()
6 ml_m = RandomForestClassifier()
7
8 dml_did = DoubleMLDID(dml_data,
9                       ml_g=clone(ml_g),
10                      ml_m=clone(ml_m))
11 dml_did.fit()
12 print(dml_did.summary)
```

	coef	std err	t	P> t	2.5 %	97.5 %
D	-0.033636	0.020753	-1.620774	0.105066	-0.074311	0.007039

Implementation - Default values

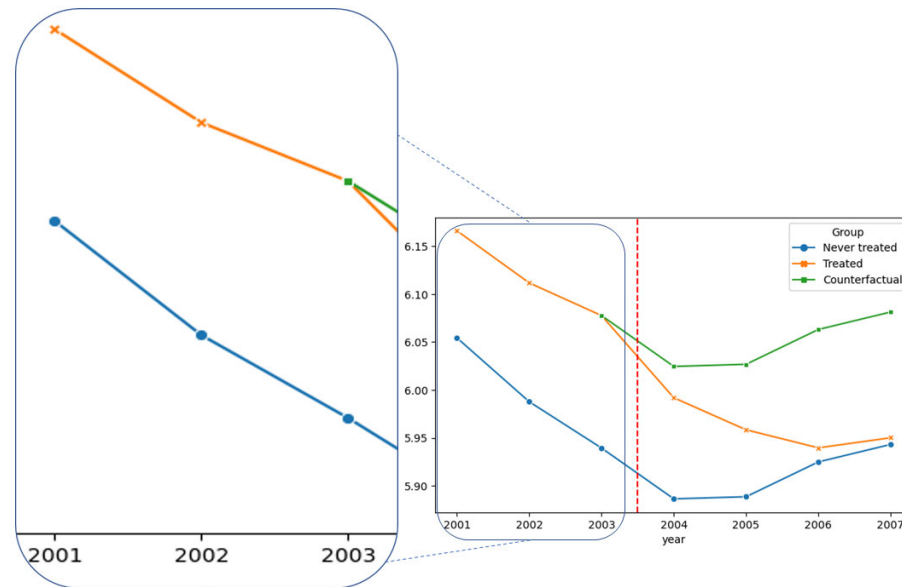
- Learner `ml_m` is only relevant for `score=observational`

```
1 from doubleml import DoubleMLDID
2 from lightgbm import LGBMClassifier, LGBMRegressor
3
4 dml_did = DoubleMLDID(dml_data,
5                       ml_g,
6                       ml_m=None,
7                       n_folds=5,
8                       n_rep=1,
9                       score='observational',
10                      in_sample_normalization=True,
11                      dml_procedure='dml2',
12                      trimming_rule='truncate',
13                      trimming_threshold=0.01,
14                      draw_sample_splitting=True,
15                      apply_cross_fitting=True)
```

Pre-Testing

Pre-Testing

- The crucial assumption for the identification of the ATTE is the conditional parallel trends assumption
- Pre-testing is a common approach to check the validity of the conditional parallel trends assumption
- If multiple time periods pre-treatment are available, the “treatment” effect can be estimated for each pre-treatment period



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Pre-Testing - Data Backend

- Select the additional pre-treatment periods from 2001 and 2002

► Code

	year	Groups	Y	lpop	lavg_pay	id	D	region_2	region_1
0	2001	Never treated	5.293305	9.771840	10.322329	13001	0	0	1
1	2002	Never treated	5.170484	9.777981	10.350063	13001	0	0	1
2	2003	Never treated	5.111988	9.790095	10.382265	13001	0	0	1
7	2001	Never treated	4.204693	8.934587	9.980124	13003	0	0	1

Pre-Testing - Estimation

- Loop over the time periods and estimate the corresponding treatment effects

▼ Code

```
1 years = [2002, 2003, 2004]
2 df_pretest = pd.DataFrame({'year': years, 'effect': np.nan, 'lower': np.nan, 'upper': np.nan})
3 for t_idx, t in enumerate([2001, 2002, 2004]):
4     df = df_subset.loc[(df_subset.year == t) | (df_subset.year == 2003)].copy()
5     df_diff = df.groupby("id").agg({"lpop": "first", "lavg_pay": "first",
6                                   "region_2": "first", "region_3": "first",
7                                   "region_4": "first", "D": "first",
8                                   "Y": compute_difference}).reset_index()
9     dml_data = DoubleMLDIDData(df_diff, y_col="Y", d_cols="D",
10                               x_cols=["lpop", "lavg_pay", "region_2", "region_3", "region_4"])
11     dml_did = DoubleMLDID(dml_data,
12                           ml_g=clone(ml_g),
13                           ml_m=clone(ml_m))
14     dml_did.fit()
15     df_pretest.loc[t_idx, 'effect'] = dml_did.coef
16     df_pretest.loc[t_idx, 'lower'] = dml_did.confint()['2.5 %'].iloc[0]
17     df_pretest.loc[t_idx, 'upper'] = dml_did.confint()['97.5 %'].iloc[0]
```

Pre-Testing - Results

- Let's take a look at the results

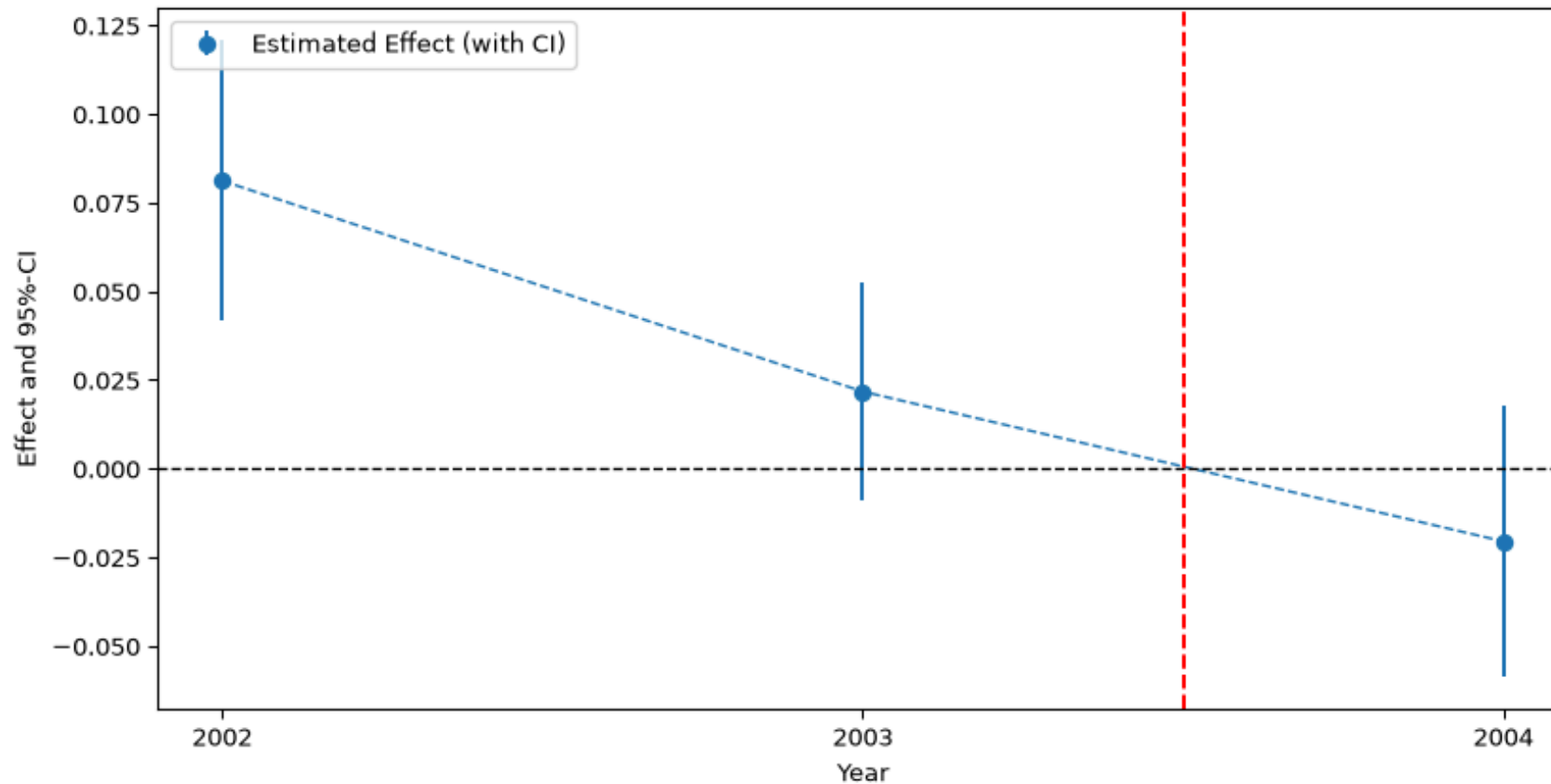
► Code

	year	effect	lower	upper
0	2002	0.081140	0.041552	0.120728
1	2003	0.021722	-0.009169	0.052614
2	2004	-0.020534	-0.058789	0.017721

Pre-Testing - Results

- It is always helpful to visualize the estimates

► Code



Multiple time periods

- We can include multiple time periods in the same manner

► Code

	year	Groups	Y	lpop	lavg_pay	id	D	region_2	region_1
0	2001	Never treated	5.293305	9.771840	10.322329	13001	0	0	1
1	2002	Never treated	5.170484	9.777981	10.350063	13001	0	0	1
2	2003	Never treated	5.111988	9.790095	10.382265	13001	0	0	1
3	2004	Never treated	5.308268	9.792891	10.424986	13001	0	0	1

Multiple time periods

- Loop over the time periods from 2005 to 2007 and estimate the corresponding treatment effects

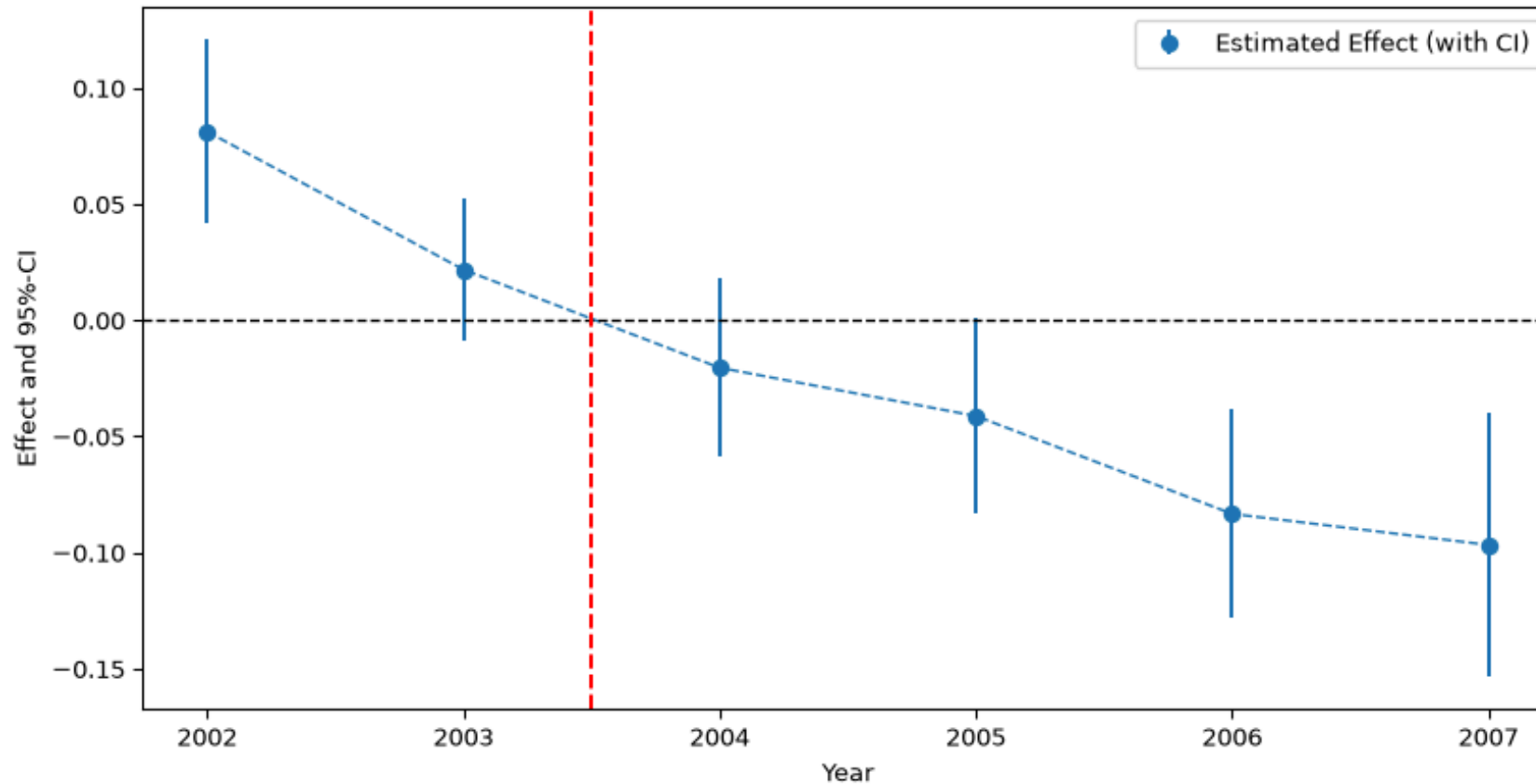
► Code

	year	effect	lower	upper
0	2005	-0.041293	-0.083163	0.000577
1	2006	-0.083440	-0.128205	-0.038676
2	2007	-0.096804	-0.153637	-0.039971

Multiple time periods - Results

- Add the pre-testing results to the plot

► Code



References

References

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- Sant’Anna, Pedro HC, and Jun Zhao. 2020. “Doubly Robust Difference-in-Differences Estimators.” *Journal of Econometrics* 219 (1): 101–22.
- Zimmert, Michael. 2018. “Efficient Difference-in-Differences Estimation with High-Dimensional Common Trend Confounding.” *arXiv Preprint arXiv:1809.01643*.

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